Project Design Phase: Solution Architecture

Date:	30 september 2026
Team ID:	[SWTID-2026-3230]
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	4 Marks

Solution Architecture Overview

Solution architecture bridges the gap between business problems and technology solutions [cite: 22]. For the 'Auto Ticket Classification using Flow Designer' project, the architecture defines the structure, behavior, and data flow of the automated ServiceNow IT service management system.

Architecture Components & Data Flow

Component Layer	Architecture Element	Description & Data Flow
Presentation Layer	ServiceNow Service Portal & UI	Provides the user interface for students and faculty to submit support tickets with short descriptions, detailed descriptions, and attachments.
Integration & Trigger Layer	Database Insert Trigger / Business Rule	Detects newly created incident records in the Task database and asynchronously fires the Flow Designer execution engine.
Processing & Triage Layer	Flow Designer & Decision Tables	Executes automated text parsing, keyword extraction, and decision matrix matching against incident fields to predict category, subcategory, and priority.
Routing & Storage Layer	Assignment Engine & CMDB Database	Automatically assigns the ticket to the correct technical assignment group (e.g., Network, Hardware, Software)

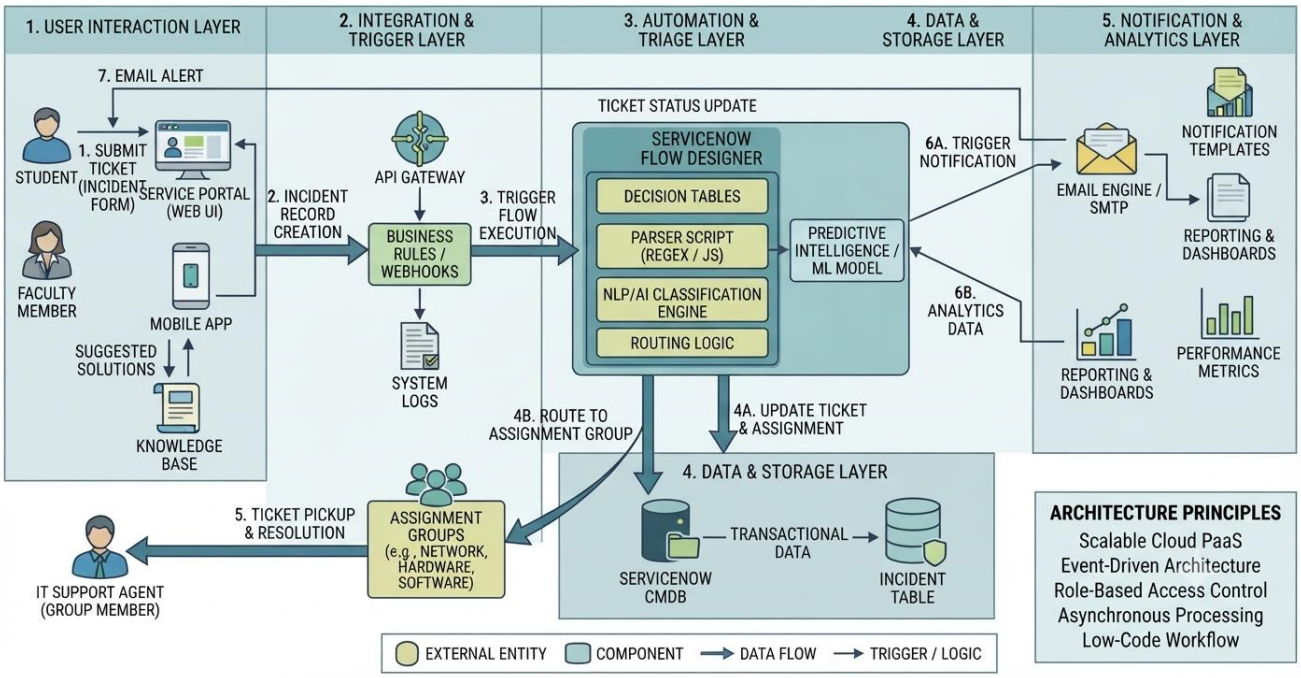
		and persists data in ServiceNow CMDB.
Notification & Logging Layer	Email Engine & CloudWatch / System Logs	Dispatches automated email notifications to end-users and support teams while logging transaction states for auditing and performance monitoring.

Solution Architecture Data Flow Diagram (Structured Representation)

The following flow outlines the end-to-end architecture pipeline corresponding to the project data flow:

- **1. User Submission:** End-user creates an incident via Service Portal -> Submits Short & Detailed Description.
- **2. Event Trigger:** New record insert triggers ServiceNow Flow Designer workflow pipeline.
- **3. Automated Triage:** Flow Designer parses text -> Evaluates Decision Tables & Keyword Matchers.
- **4. Assignment & Routing:** System assigns Category, Priority -> Automatically routes to appropriate Assignment Group.
- **5. Notifications & Output:** Dispatches confirmation emails to user -> Updates Agent Dashboard & Audit Logs.

SOLUTION ARCHITECTURE: AUTO TICKET CLASSIFICATION USING FLOW DESIGNER



ARCHITECTURE PRINCIPLES

Scalable Cloud PaaS
Event-Driven Architecture
Role-Based Access Control
Asynchronous Processing
Low-Code Workflow